

METIMMOX-1 Expected Value of Partially Perfect Information Analysis

Multi-Scenario Analysis for Biomarker-Guided Strategies in Metastatic MSS/pMMR Colorectal Cancer

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1 Summary

This report presents the results of an Expected Value of Partially Perfect Information (EVPPI) analysis for the METIMMOX-1 decision-analytic model. The analysis evaluates the value of reducing uncertainty in individual model parameters across multiple scenarios with varying willingness-to-pay (WTP) thresholds and nivolumab drug costs.

The Expected Value of Partially Perfect Information (EVPPI) quantifies the value of eliminating uncertainty in specific model parameters. It represents the maximum amount that should be paid for additional research to perfectly inform a particular parameter. EVPI (Expected Value of Perfect Information) represents the value of eliminating all uncertainty, and EVPPI for any parameter cannot exceed total EVPI.

Table 1: Analysis scenario definitions. Higher WTP thresholds (€100,000 and €150,000) are arbitrary exploratory thresholds, though the highest severity adjustment in Norwegian guidelines may fall between these values. The decreased nivolumab cost (€4,641) represents an 80% discount compared to the 40% discount in the base case scenario.

Scenario Name	WTP Threshold	Nivolumab Cost
Base Case	€ 51,000	€13,923
Decreased Nivolumab Cost	€ 51,000	€ 4,641
WTP €100,000	€100,000	€13,923
WTP €100,000 + Decreased Nivolumab	€100,000	€ 4,641
WTP €150,000	€150,000	€13,923
WTP €150,000 + Decreased Nivolumab	€150,000	€ 4,641

2 Probabilistic Sensitivity Analysis Results

2.1 Incremental Cost-Effectiveness Scatter Plot

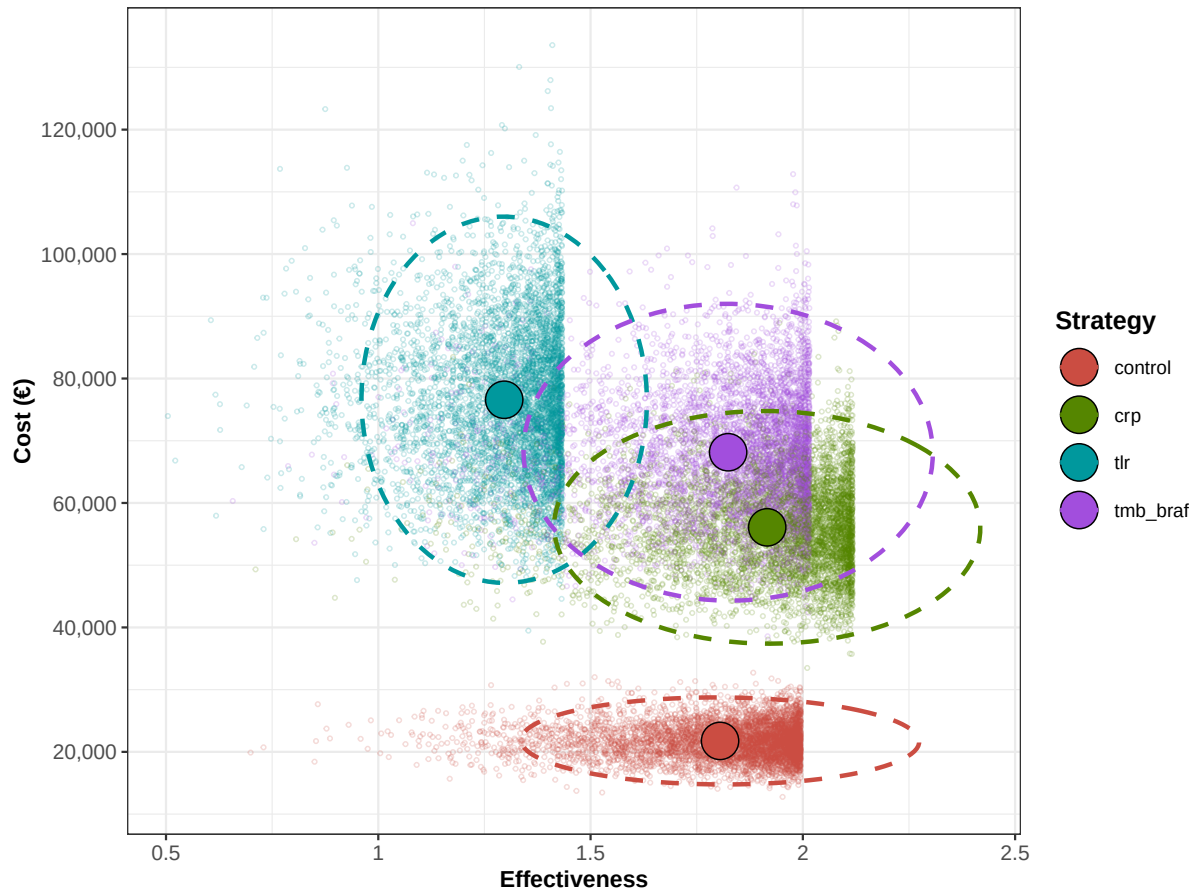


Figure 1: Incremental cost-effectiveness scatter plot from probabilistic sensitivity analysis (base case)

2.2 Cost-Effectiveness Analysis Summary

Table 2: Cost-effectiveness analysis summary from probabilistic sensitivity analysis

Strategy	Mean Cost (€)	Mean Effect (QALYs)	SD Cost	SD Effect
control	€21,758.31	1.805	€ 2,856.78	0.192
crp	€56,086.29	1.916	€ 7,638.79	0.205
tlr	€76,586.70	1.297	€12,026.29	0.137
tmb_braf	€68,154.60	1.824	€ 9,745.35	0.197

2.3 Cost-Effectiveness Acceptability Curves

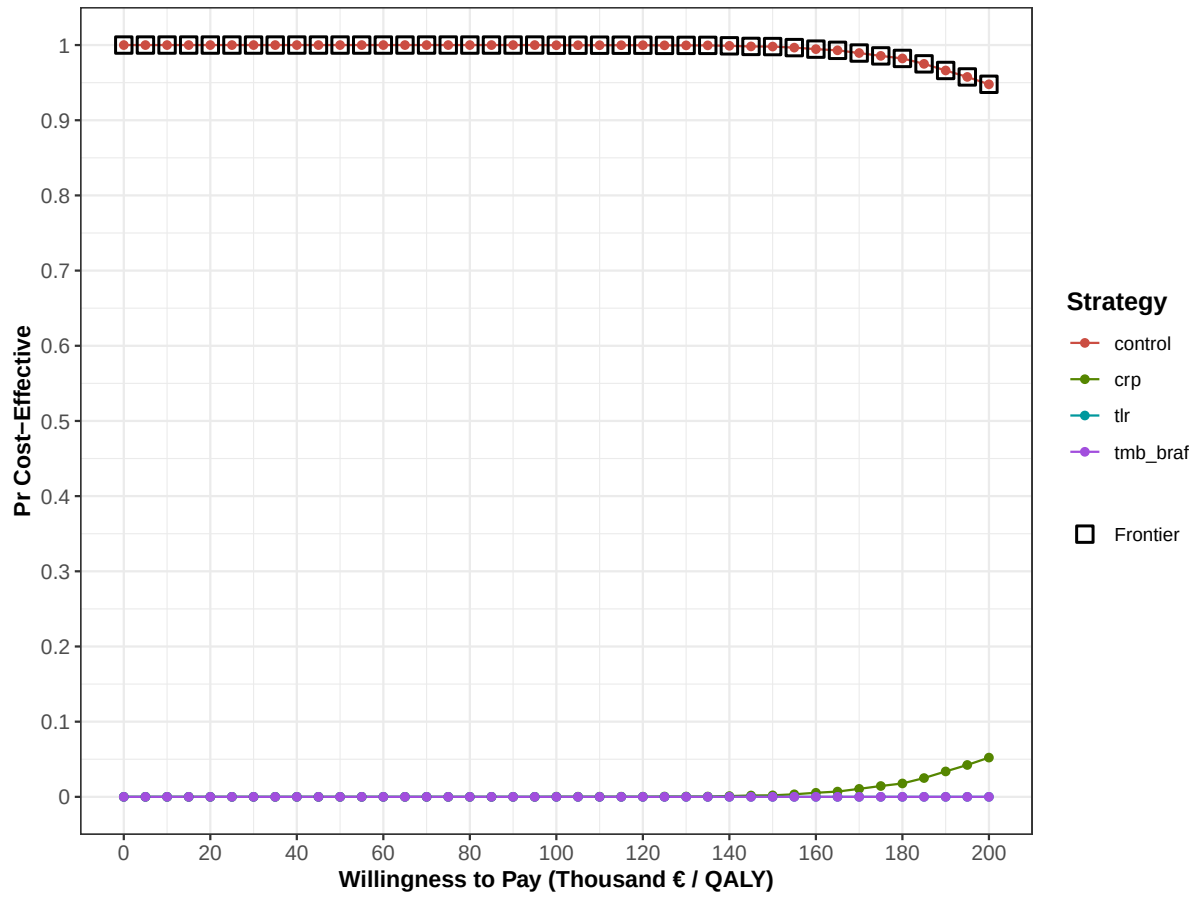


Figure 2: Cost-effectiveness acceptability curves showing probability of cost-effectiveness across willingness-to-pay thresholds

2.4 Expected Value of Perfect Information

The Expected Value of Perfect Information (EVPI) represents the maximum amount that should be paid to eliminate all parameter uncertainty in the decision model. A positive EVPI indicates decision uncertainty exists at the specified WTP threshold.

Base Case EVPI (WTP = €51,000): €0

The EVPI is effectively zero, indicating no decision uncertainty at the base case WTP threshold. In this case, one strategy is clearly dominant regardless of parameter uncertainty, making EVPI calculations less meaningful for the base case scenario.

3 Base Case Scenario

Willingness-to-Pay Threshold: €51,000

Nivolumab Cost: €13,923

Total EVPI: €0

3.1 EVPPI Results

No meaningful EVPPI results available for base case scenario due to zero EVPI.

4 Scenario 1: Decreased Nivolumab Cost

Willingness-to-Pay Threshold: €51,000

Nivolumab Cost: €4,641 (80% discount)

Total EVPI: €4.01

Reducing nivolumab costs creates decision uncertainty, making EVPPI analysis meaningful.

4.1 Incremental Cost-Effectiveness Results

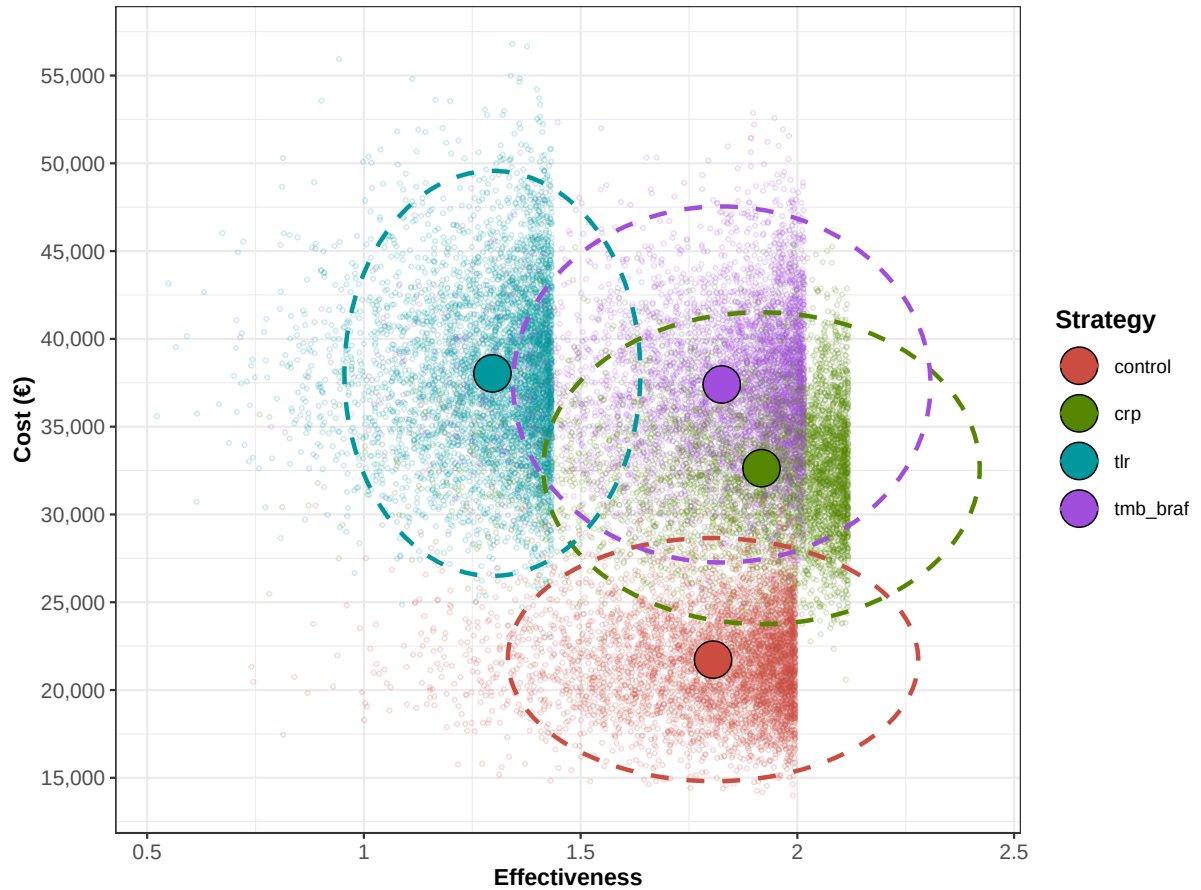


Figure 3: Incremental cost-effectiveness scatter plot (Scenario 1: Decreased nivolumab cost)

Table 3: EVPPI results: Decreased nivolumab cost

	Parameter	EVPPI	% of EVPI
1	c_drug_nivo	€4.01	100%
4	c_test_CT	€4.01	100%
5	u_np	€4.01	100%
6	u_p	€4.01	100%
7	c_other_last	€4.01	100%
9	c_other_baseline	€4.01	100%
2	c_drug_FLOX	€0	0%
3	c_test_NGS	€0	0%
8	c_other_visit	€0	0%
10	c_other_follow	€0	0%

4.2 EVPPI Results

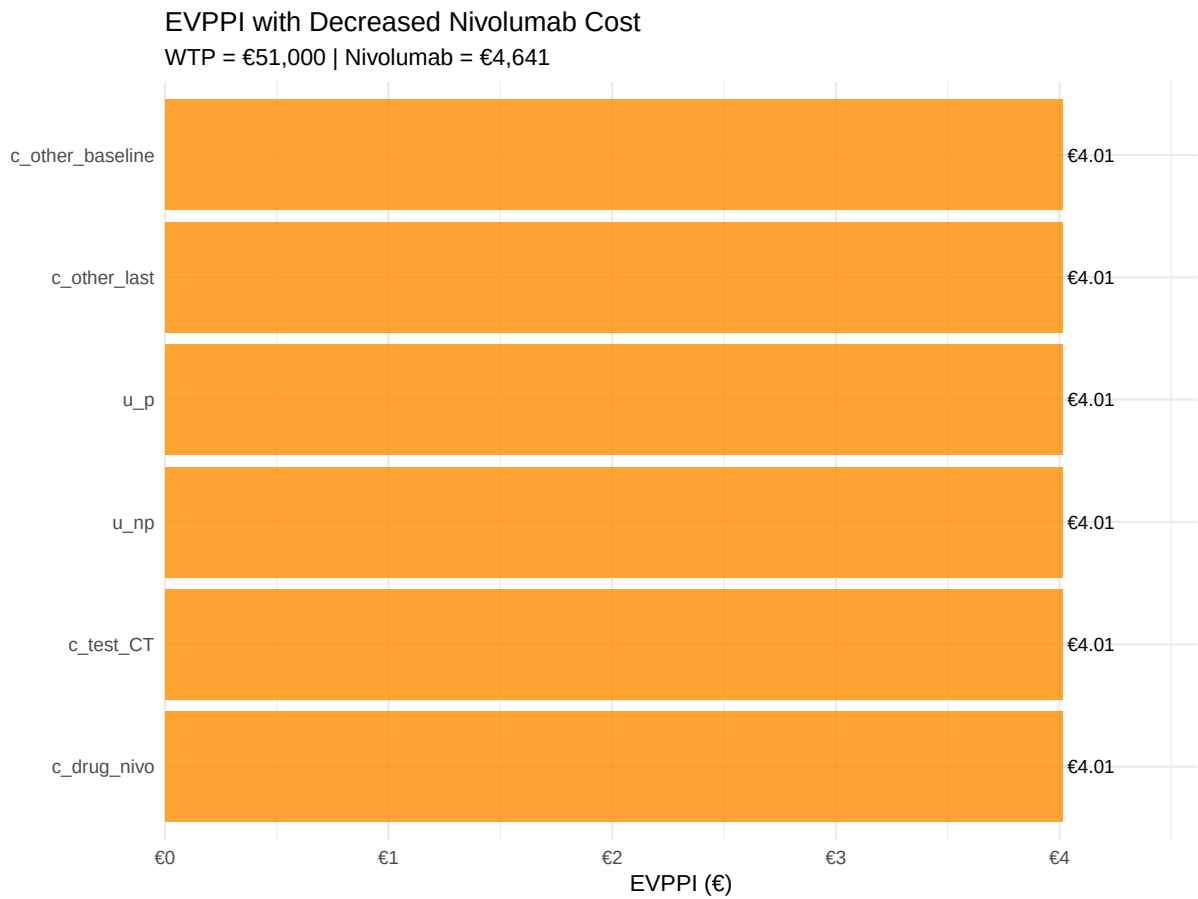


Figure 4: EVPPI with decreased nivolumab cost

5 Scenario 2: Higher WTP Threshold (€100,000)

Willingness-to-Pay Threshold: €100,000

Nivolumab Cost: Standard (base case)

Total EVPI: €0

5.1 Incremental Cost-Effectiveness Results

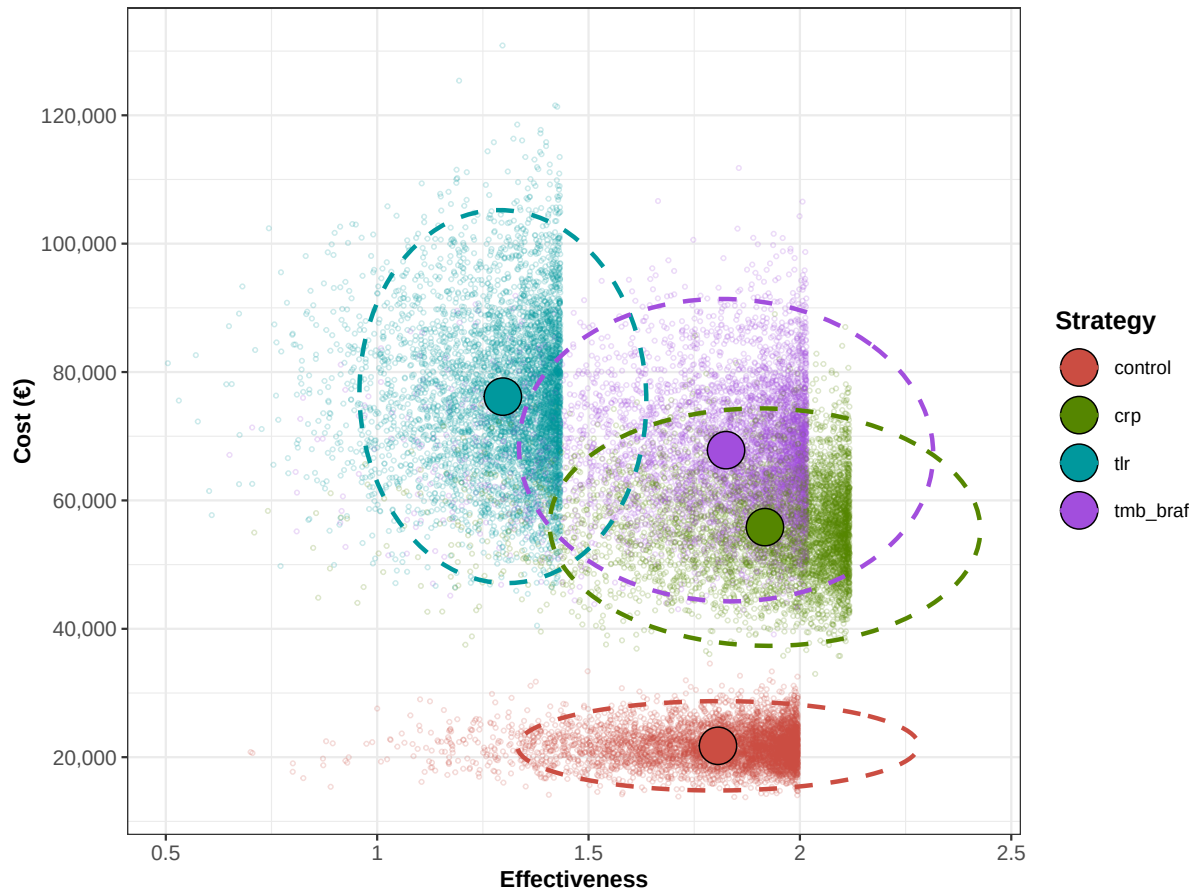


Figure 5: Incremental cost-effectiveness scatter plot (Scenario 2: WTP €100,000)

5.2 EVPPI Results

No meaningful EVPPI results available for this scenario.

6 Scenario 3: Higher WTP with Decreased Nivolumab Cost

Willingness-to-Pay Threshold: €100,000

Nivolumab Cost: €4,641 (80% discount)

Total EVPI: €1081.33

6.1 Incremental Cost-Effectiveness Results

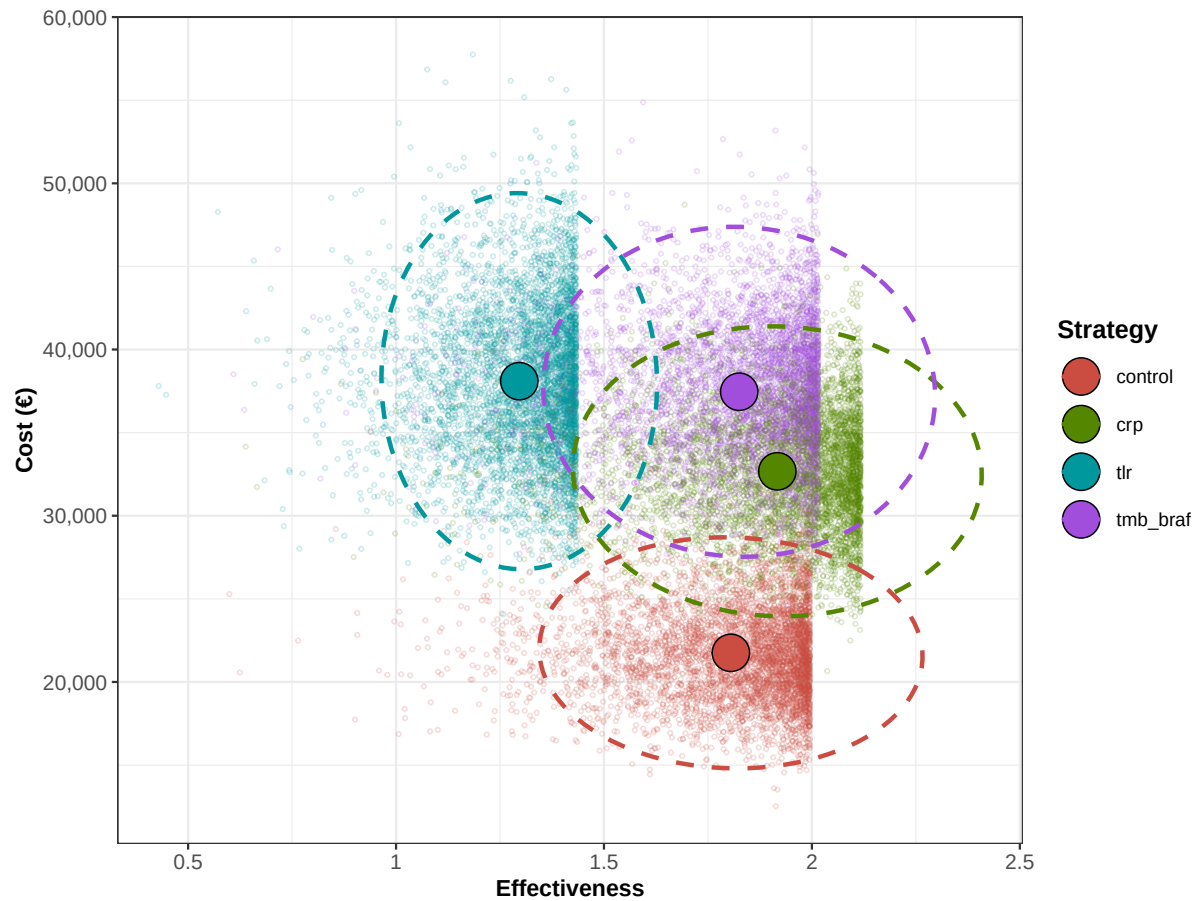


Figure 6: Incremental cost-effectiveness scatter plot (Scenario 3: WTP €100,000 with decreased nivolumab)

Table 4: EVPPI results: WTP €100,000 + decreased nivolumab

	Parameter	EVPPI	% of EVPI
4	c_test_CT	€1081.33	100%
5	u_np	€1081.33	100%
9	c_other_baseline	€1081.33	100%
8	c_other_visit	€942.17	87.1%
6	u_p	€736.93	68.1%
1	c_drug_nivo	€521.3	48.2%
10	c_other_follow	€16.59	1.5%
2	c_drug_FLOX	€0	0%
3	c_test_NGS	€0	0%
7	c_other_last	€0	0%

6.2 EVPPI Results

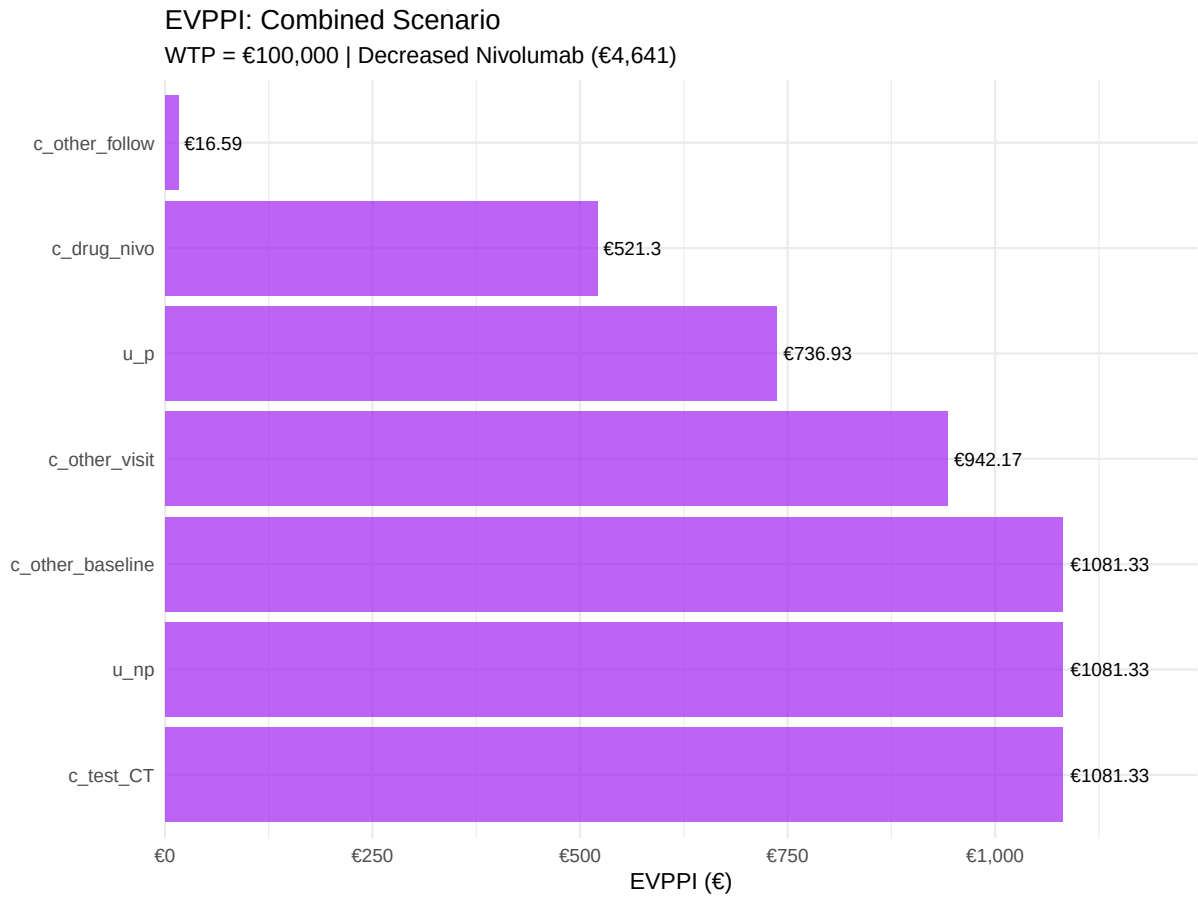


Figure 7: EVPPI: WTP €100,000 with decreased nivolumab

7 Scenario 4: Highest WTP Threshold (€150,000)

Willingness-to-Pay Threshold: €150,000

Nivolumab Cost: Standard (base case)

Total EVPI: €6.02

7.1 Incremental Cost-Effectiveness Results

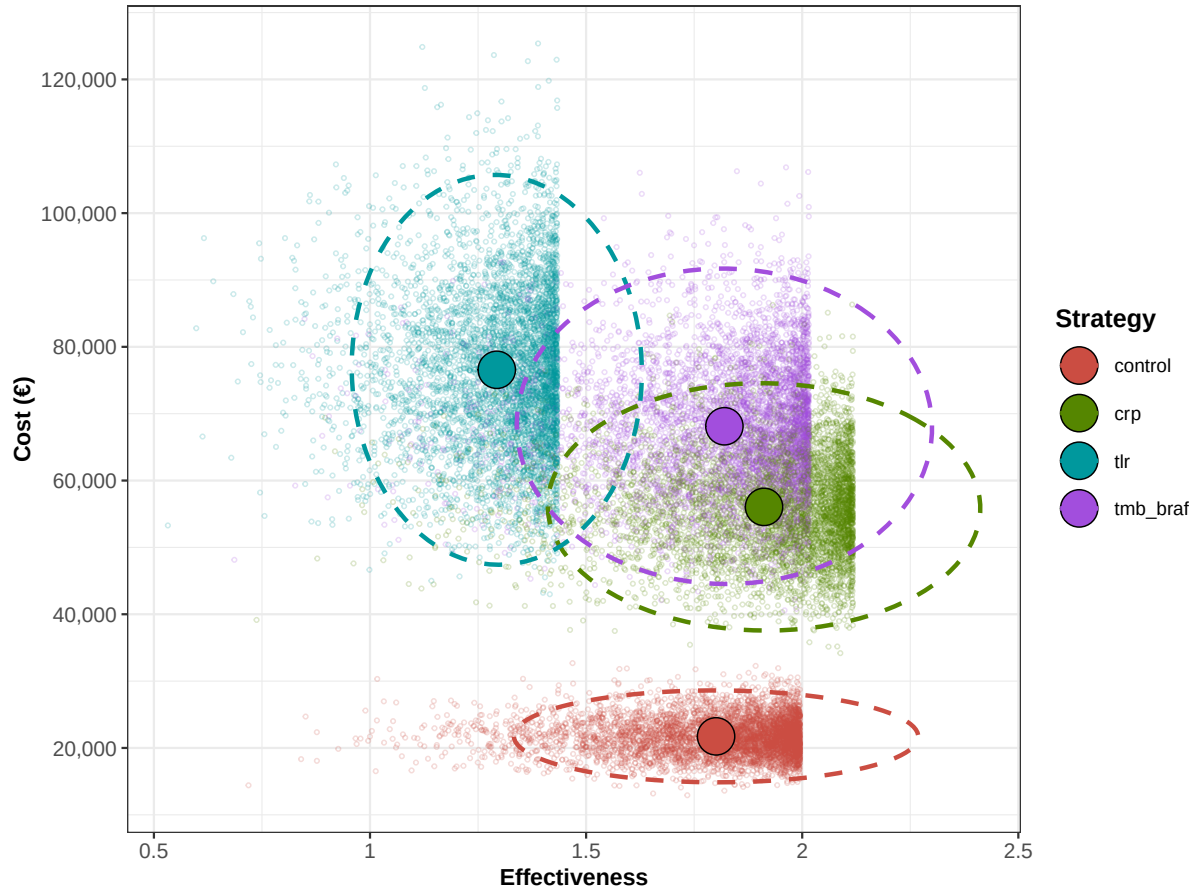


Figure 8: Incremental cost-effectiveness scatter plot (Scenario 4: WTP €150,000)

7.2 EVPPI Results

Table 5: EVPPI results: WTP €150,000

	Parameter	EVPPI	% of EVPI
1	c_drug_nivo	€6.02	100%
3	c_test_NGS	€6.02	100%
4	c_test_CT	€6.02	100%
5	u_np	€6.02	100%
6	u_p	€6.02	100%
7	c_other_last	€6.02	100%
2	c_drug_FLOX	€0	0%
8	c_other_visit	€0	0%
9	c_other_baseline	€0	0%
10	c_other_follow	€0	0%

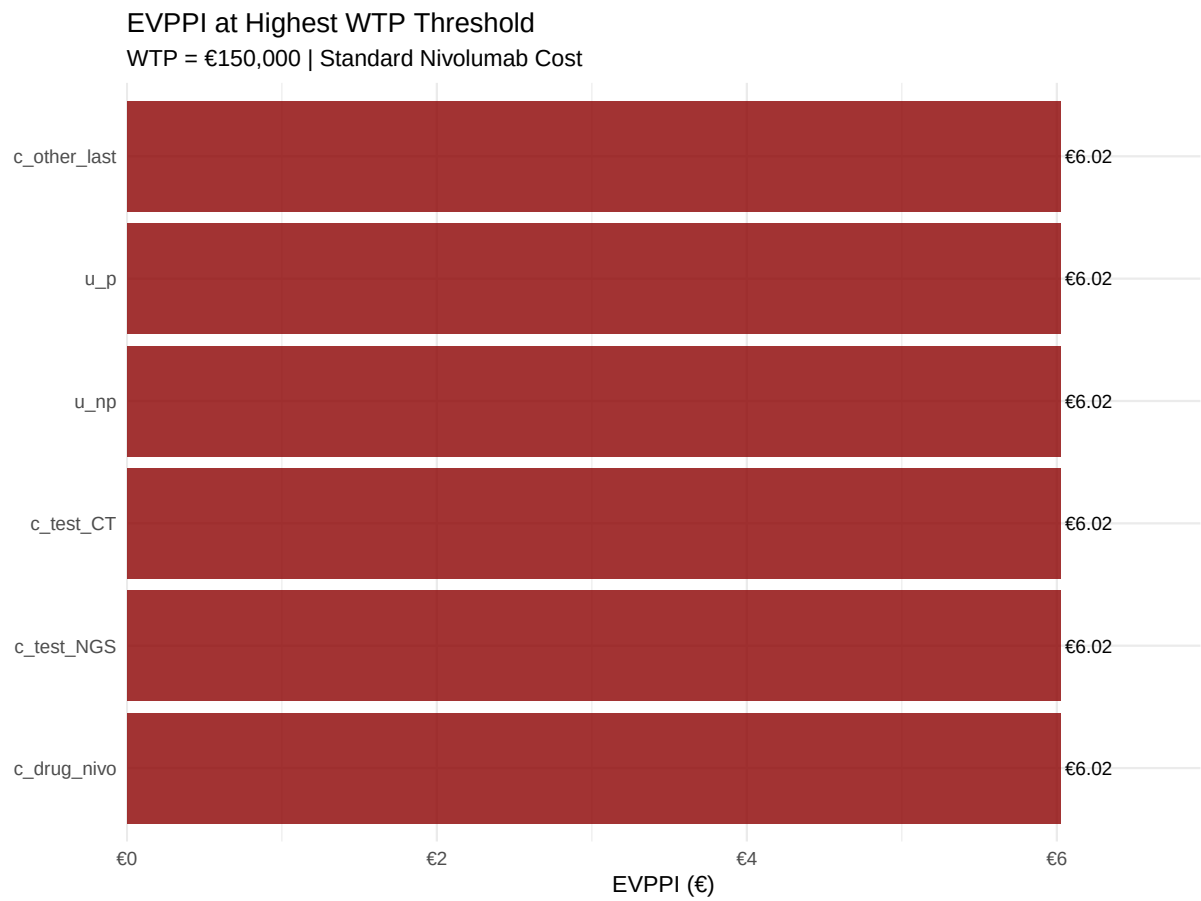


Figure 9: EVPPPI at WTP €150,000

8 Scenario 5: Highest WTP with Decreased Nivolumab Cost

Willingness-to-Pay Threshold: €150,000

Nivolumab Cost: €4,641 (80% discount)

Total EVPI: €273.11

8.1 Incremental Cost-Effectiveness Results

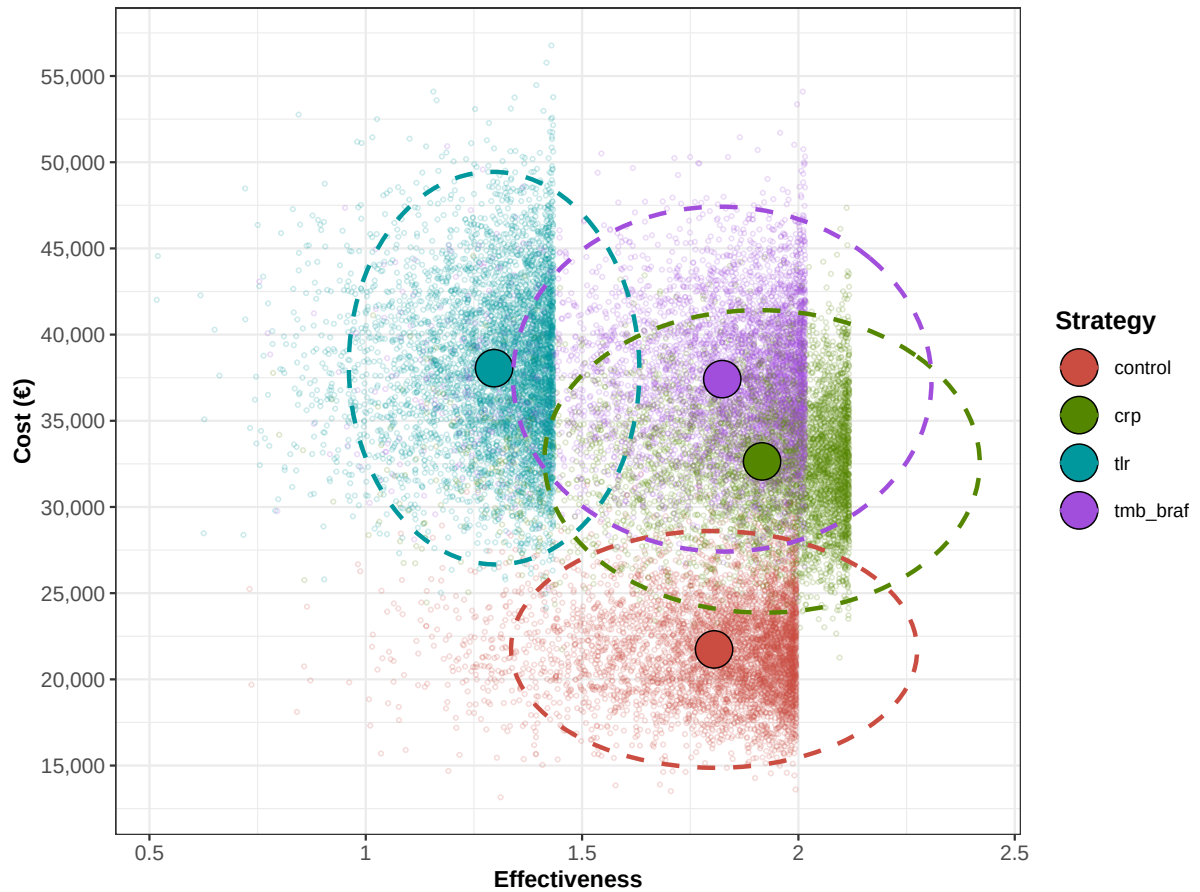


Figure 10: Incremental cost-effectiveness scatter plot (Scenario 5: WTP €150,000 with decreased nivolumab)

Table 6: EVPPI results: WTP €150,000 + decreased nivolumab

	Parameter	EVPPI	% of EVPI
1	c_drug_nivo	€273.11	100%
2	c_drug_FLOX	€273.11	100%
3	c_test_NGS	€273.11	100%
5	u_np	€273.11	100%
6	u_p	€273.11	100%
7	c_other_last	€273.11	100%
9	c_other_baseline	€273.11	100%
4	c_test_CT	€0	0%
8	c_other_visit	€0	0%
10	c_other_follow	€0	0%

8.2 EVPPI Results

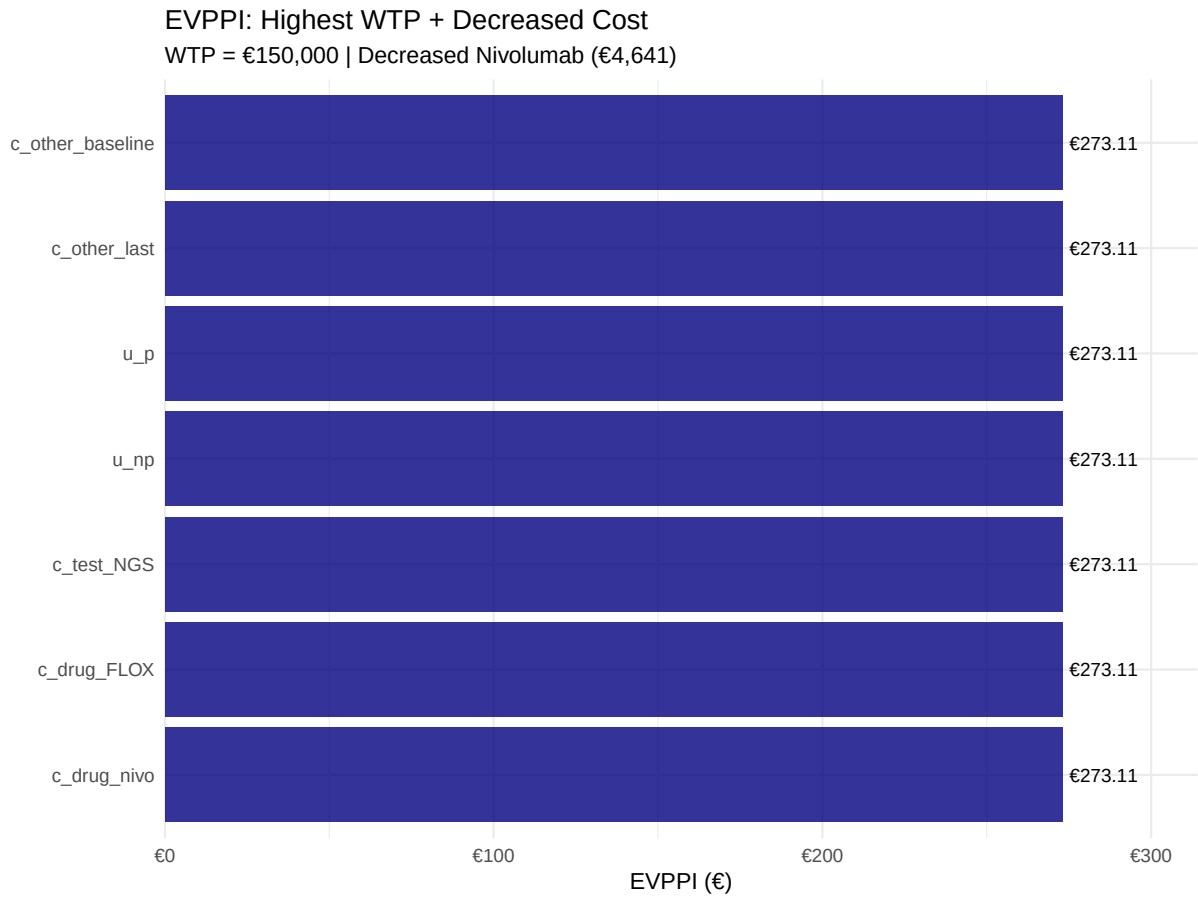


Figure 11: EVPPI: Highest WTP with decreased nivolumab cost

9 Comparative Analysis Across Scenarios

Table 7: Summary of EVPI and top EVPPI by scenario

Scenario	WTP	EVPI	Top Parameter	Max EVPPI	% of EVPI
Decreased Nivolumab Cost	€ 51,000	€4.01	c_drug_nivo	€4.01	100
WTP €100,000 + Decreased Nivolumab	€100,000	€1081.33	c_test_CT	€1081.33	100
WTP €150,000	€150,000	€6.02	c_drug_nivo	€6.02	100
WTP €150,000 + Decreased Nivolumab	€150,000	€273.11	c_drug_nivo	€273.11	100

9.1 EVPI Comparison



Figure 12: Expected Value of Perfect Information across scenarios

9.2 Parameter Importance Heatmap

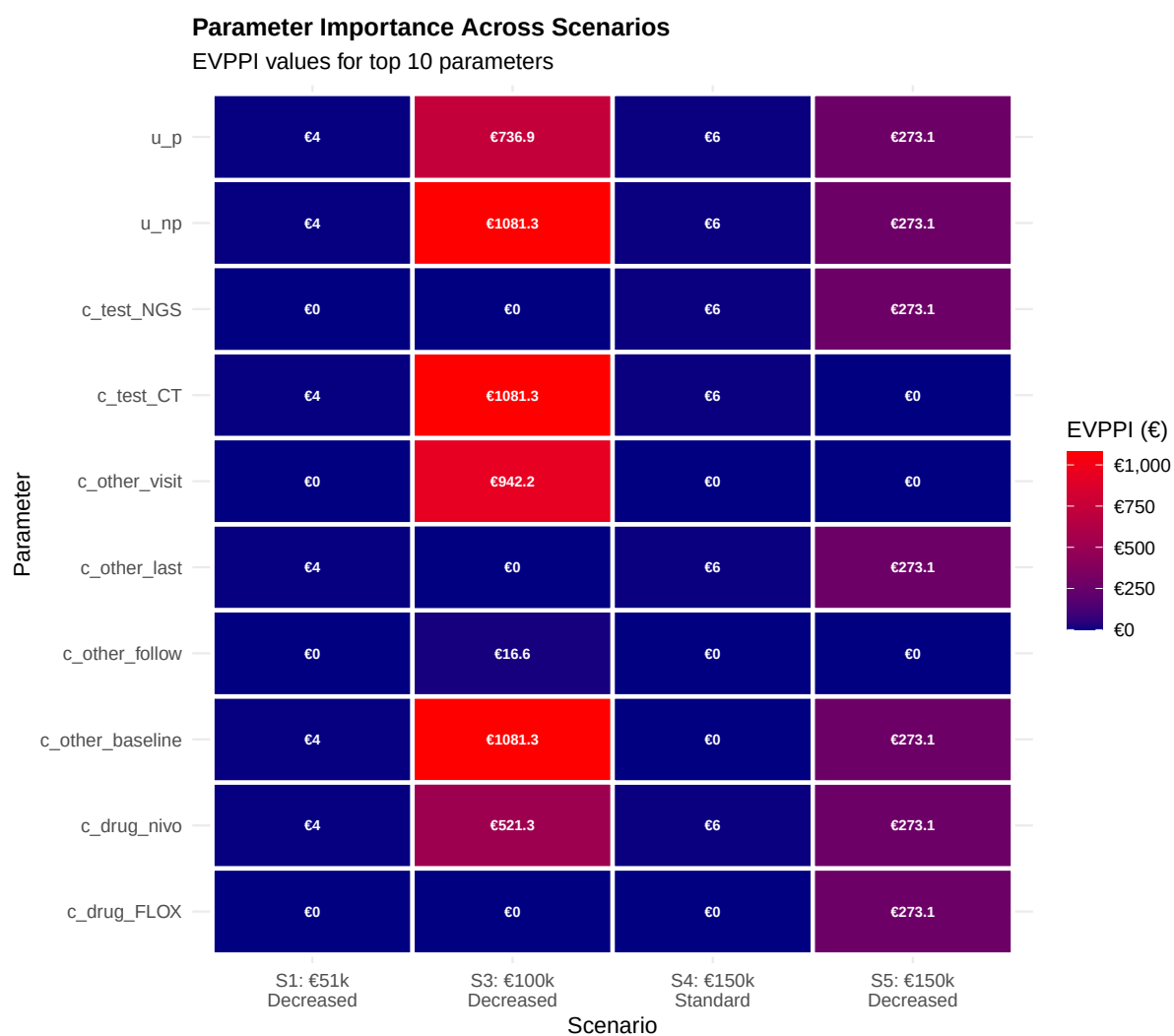


Figure 13: Parameter importance across scenarios (top 10 parameters)

10 Conclusions and Recommendations

10.1 Summary of Findings

Scenario with Greatest Decision Uncertainty: WTP €100,000 + Decreased Nivolumab (EVPI = €1081.33)

Most Important Parameter Across All Scenarios: u_{np} (Total EVPPI = €1364.48)

Impact of WTP Threshold: Higher WTP thresholds generally increase decision uncertainty (EVPI), creating greater value for additional research to reduce parameter uncertainty.

Impact of Nivolumab Cost: Decreased nivolumab costs shift the cost-effectiveness landscape and may change which parameters drive decision uncertainty.

10.2 Research Priorities

Based on the EVPPI analysis, future research should prioritize:

Table 8: Top 5 research priorities based on EVPPI

Priority	Parameter	Max EVPPI	Avg EVPPI	Relevant Scenarios
1	c_other_baseline	€1081.33	€339.61	3
2	c_test_CT	€1081.33	€272.84	3
3	u_{np}	€1081.33	€341.12	4
4	c_other_visit	€942.17	€235.54	1
5	u_p	€736.93	€255.02	4

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